THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 1: **Exploring Substances: Acidic, Basic, and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Substances that show different colours in acidic and basic solutions are called: 1

(a) catalysts **(b)** indicators
(c) reagents of taste **(d)** solvents

2. Which of these is basic? 1

(a) Soap solution **(b)** Amla juice **(c)** Tamarind water **(d)** Vinegar

3. The acid present in tamarind is: 1

(a) citric acid **(b)** lactic acid **(c)** acetic acid **(d)** tartaric acid

4. Which of these gives relief from an ant sting? 1

(a) Tamarind pulp **(b)** Baking soda paste
(c) Lemon juice **(d)** Vinegar

5. The reaction between an acid and a base is called: 1

(a) combustion **(b)** rusting **(c)** neutralisation **(d)** evaporation

6. To neutralise acidic soil, farmers add: 1

(a) vinegar **(b)** sugar
(c) common salt **(d)** lime (slaked lime)

7. Which of these is an olfactory indicator? 1

(a) Turmeric **(b)** Red cabbage **(c)** Onion **(d)** Litmus

8. Baking soda solution is: 1

(a) basic **(b)** acidic **(c)** neutral **(d)** sour

9. A solution turns blue litmus red and does not change red litmus. It is: 1

(a) a salt **(b)** acidic **(c)** basic **(d)** neutral

10. Acid rain is harmful because it: 1

(a) has no effect **(b)** is neutral
(c) helps crops always **(d)** damages buildings and water bodies

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What are acid-base indicators? Name two natural indicators. 2

12. How would you test whether a solution is acidic using litmus paper? 2

13. State two properties of acids and two of bases. 2

14. What is neutralisation? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to classify lemon juice, soap solution and sugar solution using litmus. 3

16. Explain how to prepare red rose extract and use it as an indicator. 3

17. What is neutralisation? Describe the experiment with lemon juice and lime water. 3

OR

Give three examples of neutralisation in daily life. 3

18. How can we make a household olfactory indicator and what do we observe? 3

OR

Explain why a bee sting is treated differently from a wasp sting. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Define an indicator. 5
(ii) Describe how you would prepare turmeric paper.
(iii) What colours does it show with an acid and a base?

OR

Case study: At a science fair, a volunteer sprayed a liquid on a white sheet of paper and a message appeared. 5
(i) What could the liquid be?
(ii) Why did colour appear?
(iii) Name two natural indicators.
(iv) How can you make a secret message with turmeric?

20. (i) What is neutralisation? 5
(ii) Describe an experiment to show it.
(iii) Give its uses in agriculture and health.

OR

Case study: A student feels a burning pain in the stomach after a spicy meal. 5
(i) What causes it?
(ii) What can relieve it?
(iii) Why is lemon juice a bad idea?
(iv) How does the body neutralise acid in the intestine?

END OF QUESTION PAPER — Total: 40 Marks

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